

**Technical data**

Housing:	mobile housing with carrying handles
Dimensions:	536 mm x 453 mm x 430 mm (w x h x d)
Weight:	approx. 34 kg (depending on fitments)
Protection degree:	- in case of closed housing cover: IP 54 - in case of opened housing cover: IP 31
Measuring methods:	- bi-frequency measuring method (NO ₂ , SO ₂ , H ₂ O, CO ₂) - gas filter correlation (CO, NO, HCl, NH ₃ , N ₂ O, CH ₄) - zirconium dioxide cell (O ₂)
Number of measuring components:	max. 10 infrared components (dependent on application) and oxygen
Ambient temperature:	- operation: 0...45 °C (temperature stability max. ± 5 °C) - storage: 5...35 °C (temperature stability max. 3 K/h)
Relative humidity:	max. 90% (non-condensing)
Accuracy:	< 2% of the respective measuring range
Zero point correction:	automatical with ambient air
Sensitivity correction:	with test gas, once in 6 months (sensitivity tests as standard with a concentration of 80% of the measuring range)
Cross-sensitivity correction:	additive, multiplicative
Air pressure correction:	yes
Standardisation:	dry, wet
Gas conveyance:	pump
Heat-up phase:	ready for operation after approx. 90 min (at ambient temperature of approx. 20 °C)
Media temperature:	max. 200 °C
Photometer:	- spectral range: 1...16 µm - gas path: continuously heated, standard 200 °C (higher temperatures on request) - path length of measuring cell: adjustable 2...10 m - dead volume of measuring cell: < 1 l - particle filter: 2 µm
Display/operating:	user software (MCA14m_HID.exe) via USB connection, language selectable by software (German, English, Chinese)
Data storage:	data logger function via tablet/PC
Data output:	integrated printer for output of measuring values and device configuration
Interfaces:	USB
Power consumption:	510 W
Power supply:	230 V AC, 50 Hz (optional: 115 V AC, 60 Hz)
<i>Special models are possible on request.</i>	

**MCA 14 m****Product Information**

World innovation!
Mobile hot gas analysis
without instrument air
provision

The mobile multi component analyser MCA 14 m is an extractive, continuous measuring system. It serves the continuous emission measurement of pollutants in flue gas (e.g. CO, NO, N₂O, NO₂, NH₃, CH₄, HCl, SO₂) and the measurement of CO₂, H₂O and O₂ as well as the continuous process control.

Application

The mobile MCA 14 m is applicable all-purpose for measurement of emissions, raw gases or processes. In operational emission measurement systems, amongst others, it serves the exhaust concentration control in combustion plants with different types of fuel, the thermal waste treatment, the combustion optimisation and the process management control.

Application examples:

- Power plants
- Waste incineration plants
- Refineries
- Cement industry
- Industrial exhaust air
- Paper mills
- Glass industry
- Chemical industry

**Function**

By the MCA 14 m up to ten infrared gas components can be detected simultaneously. All specification-dependent required concentrations are evaluated internally. Thereby, amongst others, the photometer control, sensor evaluation and interface communication are realised. Visualisation, operating and data logging are realised via the delivered software.

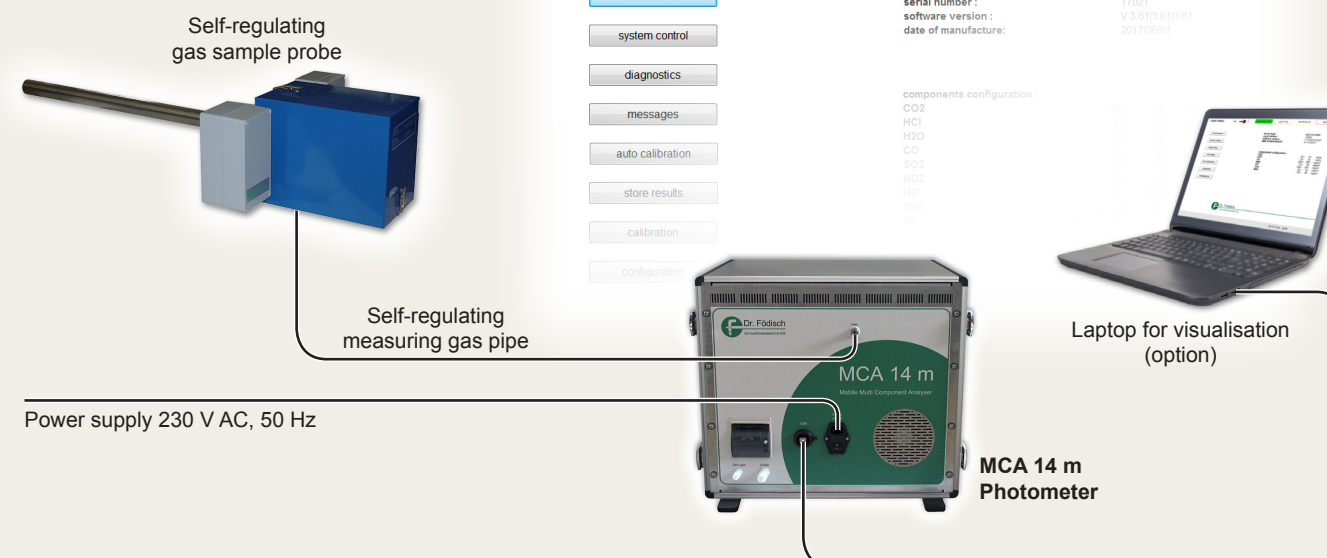
The unique characteristic of devices of this power class is that for operation of the MCA 14 m there is no instrument air supply necessary. The zero point setting is carried out with ambient air.



Measuring ranges

Component	Measuring range 1	Measuring range 2	Measuring range 3
CO	0...75 mg/m ³	0...300 mg/m ³	0...5000 mg/m ³
CO ₂	0...25 vol. %	0...50 vol. %	-
NO	0...100 mg/m ³	0...400 mg/m ³	0...3000 mg/m ³
NO ₂	0...50 mg/m ³	0...500 mg/m ³	-
N ₂ O	0...50 mg/m ³	0...3000 mg/m ³	-
NH ₃	0...10 mg/m ³	0...50 mg/m ³	0...500 mg/m ³
SO ₂	0...50 mg/m ³	0...300 mg/m ³	0...2500 mg/m ³
HCl	0...15 mg/m ³	0...90 mg/m ³	0...5000 mg/m ³
H ₂ O	0...40 vol. %	-	-
CH ₄	0...50 mg/m ³	0...500 mg/m ³	-
O ₂	0...25 vol. %	-	-

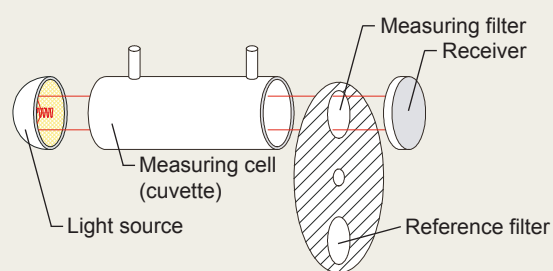
System design



Measuring methods

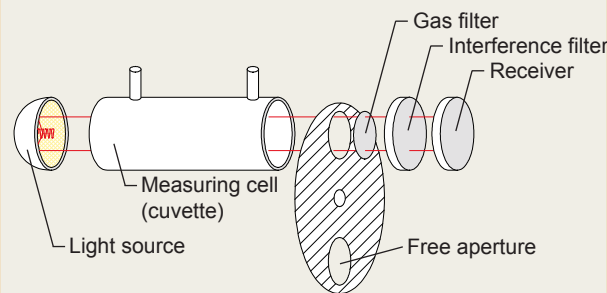
Bi-frequency measuring method

At bi-frequency measuring method a measuring filter and subsequently a reference filter are directed into the light path. Thereby the transmission range of the measuring filter lies in the range of the absorption wave length of the measuring component. The transmission range of the reference filter lies in the spectral range outside of the absorption wave length of the measuring component. So the light intensities with and without gas influence can be determined successively.



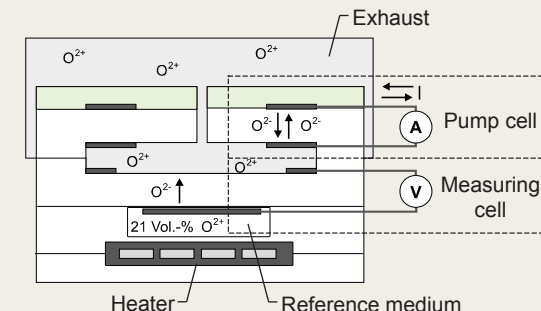
Gas filter correlation

At gas filter correlation the spectral sphere of action of the reference filter is identical with the one of the measuring gas filter. The reference filter position possesses an additional gas filter which is filled with a high concentration of the gas to be measured in order to reach the saturation of the infrared absorption. Thereby the substance-specific spectrum of the infrared light is almost completely faded out.



Zirconium dioxide cell

The oxygen measurement is carried out by means of a zirconium dioxide cell. Inside the cell the measuring gas is separated from the reference gas (ambient air) via a zirconium membrane. Depending on the oxygen partial pressure the oxygen ions move through the membrane. This results in an electric potential difference. The pump cell provides a constant oxygen concentration in the measuring cell. The thereby spent energy is a degree for the oxygen concentration.



Highlights of the device

- mobile hot gas analyser system in small format
- continuous, extractive measurement of up to ten infrared components and oxygen
- no instrument air necessary
- field-proven components, modern photometer technology
- self-sustaining operation by pump supply
- long operation times, high reliability
- easy placement directly at the measuring point
- pre-calibrated → immediately deployable
- integrated zero gas provision with ambient air
- visualisation and operating via delivered software
- integrated printer for data output
- first-class price-performance ratio